

14 (a) (i) ${}^4_2\text{He}$ or ${}^4_2\alpha$ B1 [1]

(ii) ${}_0^1\text{n}$ B1 [1]

(b) (i) $\Delta m = (29.97830 + 1.00867) - (26.98153 + 4.00260)$ C1

$$= 30.98697 - 30.98413$$

$$= 2.84 \times 10^{-3} \text{ u} \quad \text{C1 [2]}$$

(ii) $E = c^2 \Delta m$ or mc^2 C1

$$= (3.0 \times 10^8)^2 \times 2.84 \times 10^{-3} \times 1.66 \times 10^{-27}$$

$$= 4.2 \times 10^{-13} \text{ J} \quad \text{A1 [2]}$$

(c) mass of products is greater than mass of Al plus α
or

reaction causes (net) increase in (rest) mass (of the system) B1

α -particle must have at least this amount of kinetic energy B1 [2]